

TOPIC 45/60

Embedding Models in RAG

The Translators of AI.

How we convert raw human language into high-dimensional geometric maps for accurate machine retrieval.

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THE CORE CONCEPT

Language to Math

Computers do not understand text. They understand numbers. Embedding models are specialized neural networks that map human text to **vectors**.

Vector Embedding

An embedding is an array of thousands of numbers (e.g., 768 or 1536). Every dimension represents a latent feature of the text—capturing nuance, context, and synonymy that simple keyword matching misses.

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RETRIEVAL LOGIC

Keyword vs. Semantic

Keyword (Old School)

Literal matching (e.g., SQL).
If the user types "dog" but the doc says "canine," retrieval fails. It doesn't understand context.

Semantic (Modern RAG)

Contextual matching. In vector space, "dog" and "canine" are geometrically identical. The system captures the *concept*.

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THE MATH

Geometry of Meaning

The magic happens in latent space. Every piece of text is a point in a massive, multi-dimensional grid.

```
// Visualizing a point in 1536 dimensions
```

```
"The quick brown fox" → [0.01, -0.04, 0.92, ...]
```

```
"A speedy auburn mammal" → [0.01, -0.03, 0.91, ...]
```

Because these vectors are nearly identical in coordinate space, they are semantically similar.

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HOW THEY LEARN

Contrastive Learning

Embedding models are trained using "triplets" to learn which concepts belong together and which do not.

Positive Pair (Pull Together)

"King" + "Queen". Embeddings trained to minimize the distance between related concepts.

Negative Pair (Push Apart)

"King" + "Banana". Embeddings trained to maximize the distance between unrelated concepts.

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THE LANDSCAPE

Top Embedding Models

The choice of embedding model is a critical decision in any RAG architecture.

Proprietary

OpenAI (text-embedding-3-small/large), Cohere Embed v3. Highly optimized, massive training data, pay-per-token.

Open Source

BGE (Beijing Academy), E5 (Microsoft), Jina Embeddings. Often perform at parity with proprietary options, free to host.

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PERFORMANCE

Similarity Search

How do we retrieve the data? We compute the distance between the query vector (Q) and every doc vector (D) in the DB.

Cosine Similarity

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

Measuring the angle between vectors rather than the absolute magnitude, providing a scale-invariant measure of similarity.

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PRO-TIPS

The Gotchas

Domain Mismatch

An embedding model trained on generic web text will fail on medical or legal code. Use domain-specific embeddings.

Context Limits

Most embedding models have a 512 or 8192 token limit. Truncation is often required, potentially losing data.

Normalization

Always ensure vector normalization before indexing to allow fast similarity math.

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ARCHITECTURE

Separation of Concerns

Embedding
RETRIEVAL

Responsible for Finding the needle
in the haystack.

Generation
REASONING

Responsible for Reading the needle
to draft the answer.



